

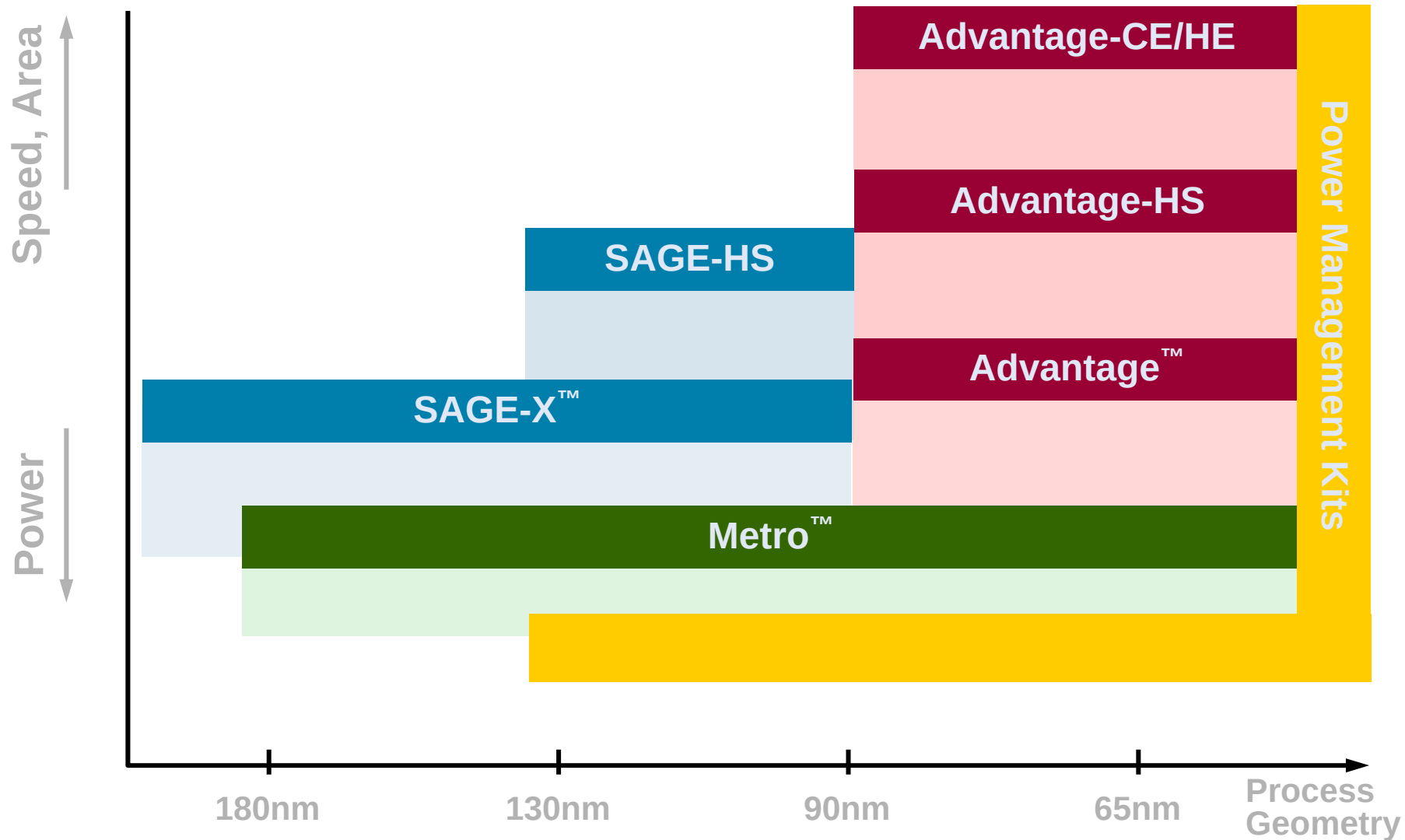


Standard Cell Products

February 2006

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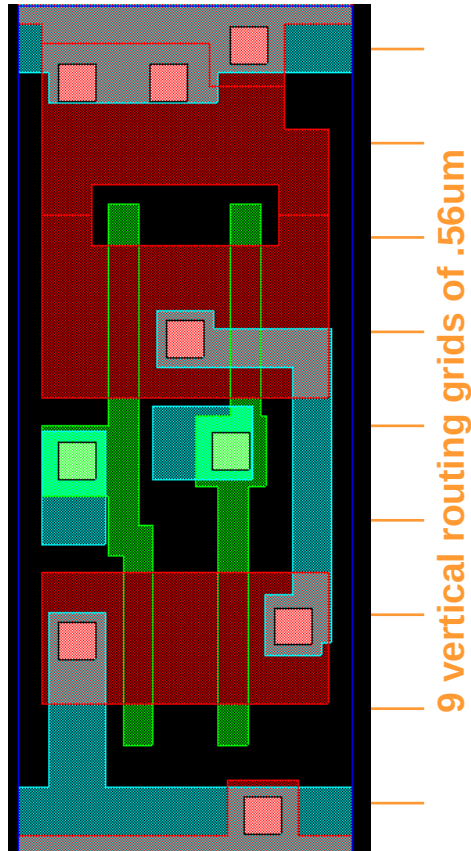
Standard Cell Families



SC Architecture – Number of Tracks

NAND2 SAGE-X 180G TSMC

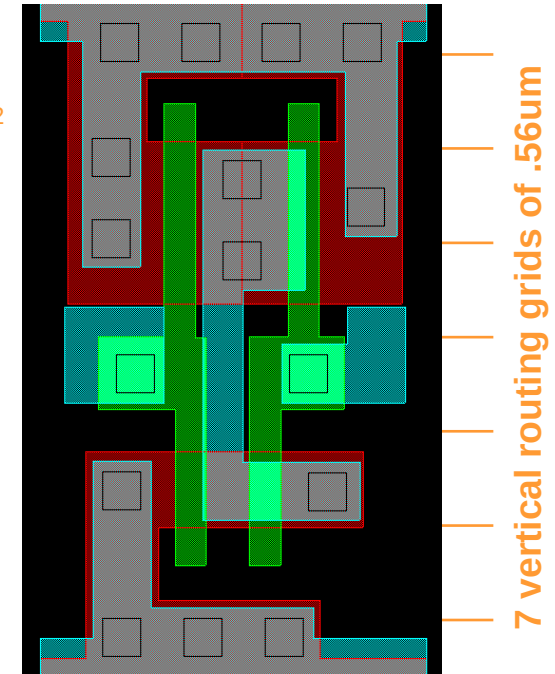
Cell size x 1.98um
Cell size y 5.04um
Total 9.98um²



- 9-track cell height for performance – density optimized cell design

NAND2 Metro 180G TSMC

Cell size x 2.24um
Cell size y 3.92um
Total 8.78um²



- Smaller cell height and overall smaller area, cells often slightly wider
- Less space for transistors → lower performance and power

Standard Cell Library Comparison

	SAGE-X	Metro	Advantage	Advantage-HS SAGE-HS	Advantage CE/HE15
Target designs	High speed, high density	Low power and high density	Higher speed with high density	High performance	Highest Performance Data path routing
Speed	1	0.75	1.1	1.25	1.3
Power	1	0.75	1.1	1.25	1.3
Area	1	0.85	1.0	1.2	1.5 (very high horizontal porosity)
# tracks high	9	7 (180 to 90nm) 8 (65nm)	9 (90nm) 9 or 10 (65nm, depending on fab)	11 (130nm) 12 (90 and 65 nm)	15
Well taps	included	separate	separate	separate	separate
Metal usage	M1 only	M1 only	M1 only	M1 only	M2 power; M1 only with two exceptions
Availability for process nodes (nm)	250 to 90	180,130, 90, 65	90 (65 in development)	130 (90 & 65 in development)	90 as beta

METRO

- Low power and high density standard cell library
 - 25% lower power and 15% less area compared to SAGE-X
 - Cells are 7 (180 to 90nm) or 8 (65nm) M1 tracks high
- Large cell set with focus on power optimization
 - 632 cells, 135 functions (180, 130, 90, 65nm)
 - Rich set of combinatorial and sequential functions
 - Integrated clock gating cells for power optimization support
 - Proprietary circuit designs optimized for low active and static power
 - Distribution of drive strengths minimizes over-design, leading to minimum active and static power
- Cells designed w/o well taps
 - Higher transistor density compared to libraries with well taps in every cell
 - Enables the use of advanced power saving techniques provided via the Power Management Kit, i.e. back-bias for leakage reduction

ADVANTAGE 90nm

- Designed for high speed by maintaining high density
 - 5 to 10% better performance compared to SAGE-X by maintaining cell size of 9 tracks and identical cell width
 - Cell set with 640 cells, 147 functions (same as SAGE-X v3.0)
- Cells designed w/o well taps
 - Available space inside cell used to grow transistor width for better performance
 - Enables the use of advanced power saving techniques provided via the Power Management Kit, i.e. back-bias for leakage reduction
- All other features similar to standard SAGE-X library

Advantage and Advantage-HS for 65nm

- Large cell set with focus on high performance via synthesis
 - Combination of Advantage-CE and SAGE-X/HS features
 - Cells are 9/10 (Advantage) or 12 (Advantage-HS) routing tracks high
 - Performance – power optimization for 12 track version
 - Performance – power – area optimization for 10 track version
- Advanced high performance feature set
 - 930 cells, 139 functions per Vt type
 - Rich set of drive strengths with fine granularity
 - Multiple beta ratios for basic cells
 - Additional combinatorial cells and faster FF's compared to SAGE-X
 - Well taps outside cells
 - Support of advance power saving features

Comprehensive Power Management

- ARM® Power Management Kits
- Support for broad range of sophisticated power management techniques
 - Active control of dynamic power
 - Active control of static / leakage power
- Available as add-on for all ARM standard cell products
 - Full feature set requires tapless library
- Close collaboration with EDA partners
 - Reference methodologies
 - Modeling
 - Automation
- Used by the ARM Intelligent Energy Management System (IEM)

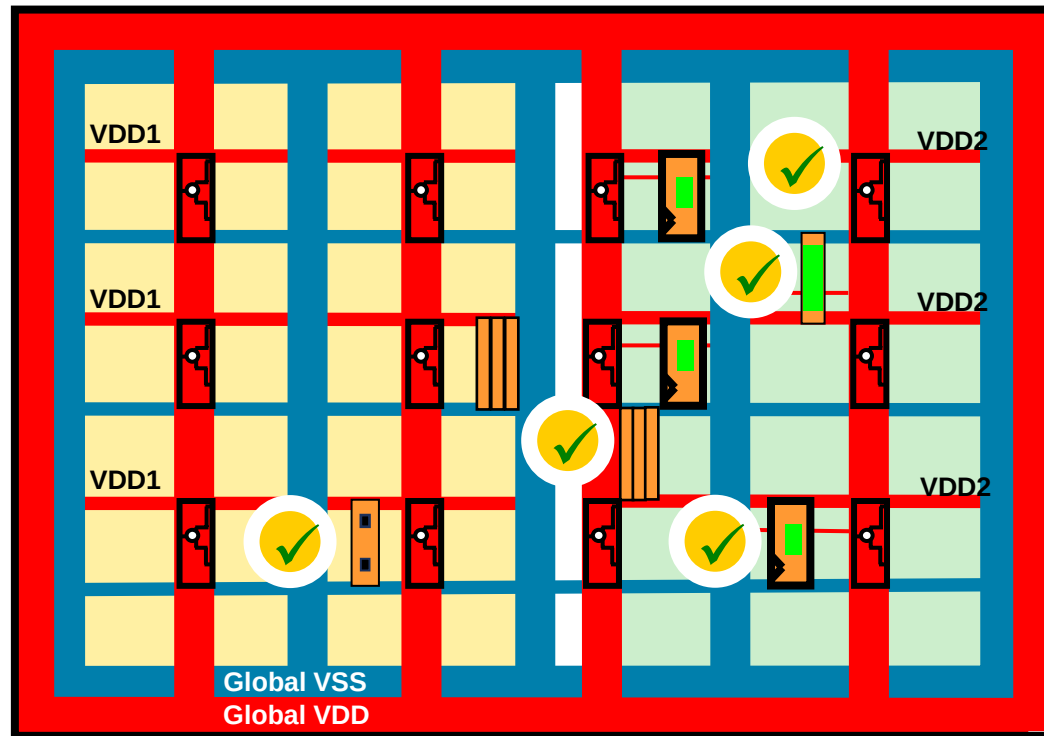


Enable IC designers to achieve 5X reduction in power

ARM Power-Management Kit

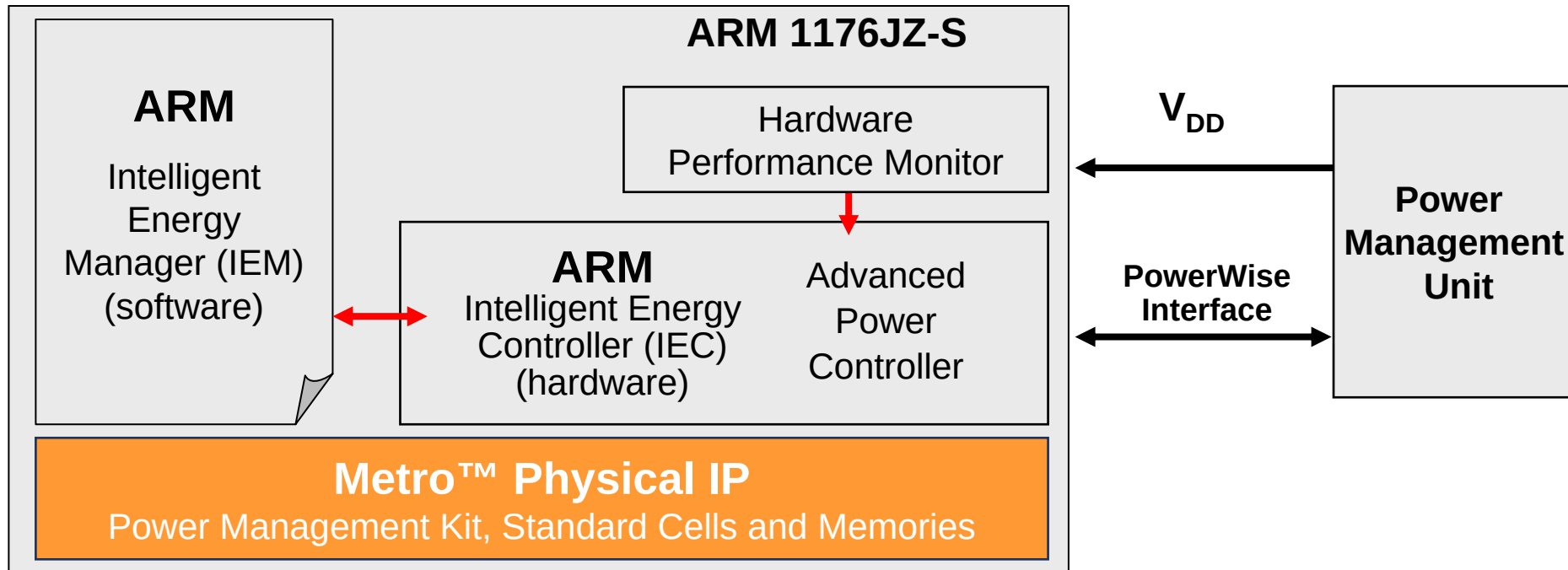
- MT-CMOS power gates*
 - Power control of voltage islands via switchable voltage rails using header (shown) or footer cells
- Level shifter and isolation cells
 - Up and down shifting with optional enable signal
- Retention flip-flops*
 - Maintain FF state after power down for leakage reduction
- Back-bias support*
 - Reduce leakage current via well-biasing with special fill_tie cells
- Always on buffer*
 - Buffering of signals in powered-down areas

*Note: requires tapless library



Note: Picture shows a conceptual implementation

ARM's Artisan™ Physical IP for ARM IEM™



- IEM methodology optimizes energy consumption with voltage and frequency scaling
- Metro technology are used on physical IP level
 - Power management Kit to support power islands with different voltage levels and power down of chip areas
 - Standard cell library and memories optimized for dynamic and leakage power
 - Standard cells and memories characterized over extended operating range, including low voltages

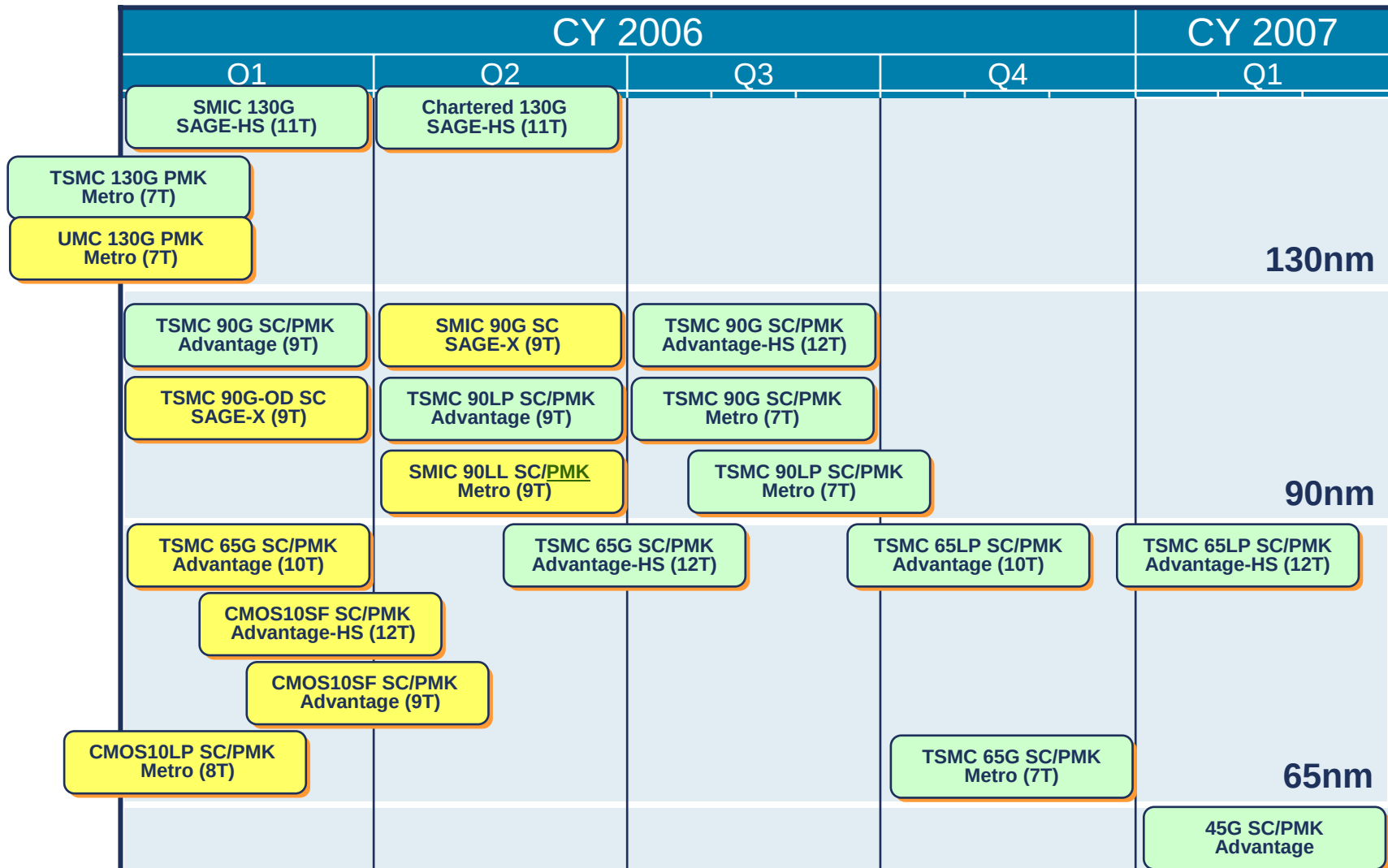
SC Availability Overview 1Q/2006

	Sage-X 9-track	Metro 7/8-track	Advantage 9/10-track	Advantage- HS (Sage-HS) 11/12-track	Advantage- HE15 15-track	PMK
65nm	-	FLP – IBM+	FLP: IBM+ FLP: TSMC	FLP: IBM+		FLP: Metro, ADV - IBM
90nm	FLP	FLP – IBM+	OTS: TSMC FLP – UMC, CSM	Planned	OTS: TSMC Beta Version	FLP: Metro-IBM OTS: Advantage TSMC
130nm	FLP	OTS: TSMC, CHSM FLP: UMC, SMIC	-	OTS: TSMC		OTS: Metro- TSMC FLB: Metro-UMC
180nm	FLP	OTS – TSMC FLP: SMIC	-	-	-	-

Available

Under development

Standard Cell Roadmap – 1Q06



Right edge of box = date of production release

Only major release are shown, depending on customer and fab priorities changes are possible

Access Library
(free)

Application Specific
(User Licensable)

SC Availability TSMC

ARM Standard Cell Product Availability for TSMC

STANDARD CELL PRODUCTS	CLN90				CL011		CL013					
	LP	G	G-OD	GT	LVP	G	G	LV	LV-OD	LVLK	LVLK-OD	LP
SAGE/SAGE-X (speed & density optimized)		StdVt, HVt (LVt OTS)	StdVt, HVt	StdVt, HVt (LVt OTS)	StdVt, HVt	StdVt, HVt	StdVt (HVt OTS)	StdVt, HVt	StdVt, HVt	StdVt, HVt	StdVt, HVt	StdVt
Metro (low power and high density)							StdVt, HVt					
Advantage (high speed & density optimized)		StdVt, HVt, LVt	StdVt, HVt, LVt	StdVt, HVt								
SAGE-HS (very high speed optimized)							StdVt	StdVt	StdVt	StdVt	StdVt	
Advantage-HS (very high speed optimized)												
Advantage-HE15 (ultra high speed for cpu cores)		StdVt, HVt, LVt										

STANDARD CELL PRODUCTS	CL015		CL018				CL025	
	G	LV	G	LV	LP	CIS	G	E
SAGE/SAGE-X (speed & density optimized)								
Metro (low power and high density)								
Advantage (high speed & density optimized)								
SAGE-HS (very high speed optimized)								
Advantage-HS (very high speed optimized)								

Explanation	
Free library program (available now)	
Free library program (in development)	
Application-specific - OTS (available now)	
Application-specific - OTS (in development)	
Not Available	

SC Availability Chartered

ARM Standard Cell Product Availability for Chartered

STANDARD CELL PRODUCTS	CMOS 90nm					CMOS 130nm				CMOS 180nm			
	9SF-AT	9SF 1.2V	9SF 1.0V	9SF 1.2V	9FLP	HP	N	HVT	LP 1.2V-1.5V	CB	IB	IB-LP	ULL
SAGE/SAGE-X (speed & density optimized)		LVt	RegVt	eMPUFET		RegVt, HVt						HVT	
Metro (low power and high density)					RegVt, HVt, LVt, SVt		RegVt, HVt						
Advantage (high speed & density optimized)	RegVt, LVt			eMPUFET									
SAGE-HS (very high speed optimized)													
Advantage-HS (very high speed optimized)													

Explanation	
Free library program (available now)	
Free library program (in development)	
Application-specific - OTS (available now)	
Application-specific - OTS (in development)	
Not Available	

Last update 31-Jan-06

SC Availability IBM

ARM Standard Cell Product Availability for IBM

STANDARD CELL PRODUCTS	CMOS 90nm				CMOS 130nm				CMOS 180nm			
	9SF 1.2V	9SF 1.0V	9SF 1.2V	9FLP	8RF-LP	8RF	8FSG	8SFGMP	7RF	7SF	IB-LP	ULL
SAGE/SAGE-X (speed & density optimized)	LVt	RegVt	eMPUFET		1.2V, 1.5V		RegVt				HVT	
Metro (low power and high density)				RegVt, HVT, LVt, SVt								
Advantage (high speed & density optimized)												
SAGE-HS (very high speed optimized)												
Advantage-HS (very high speed optimized)												

Explanation	
Free library program (available now)	
Free library program (in development)	
Application-specific - OTS (available now)	
Application-specific - OTS (in development)	
Not Available	

SC Availability UMC

ARM Standard Cell Product Availability for UMC

	L80	L90		L130E			L180		L250
STANDARD CELL PRODUCTS	G-OD	G-OD	SP-OD	SP	LL	HS	GII	UMCJ LL	
SAGE/SAGE-X (speed & density optimized)	RegVt, HVt	RegVt, HVt	RegVt, HVt						
Metro (low power and high density)									
Advantage (high speed & density optimized)		RegVt, HVt							
SAGE-HS (very high speed optimized)									
Advantage-HS (very high speed optimized)									

Explanation	
Free library program (available now)	
Free library program (in development)	
Application-specific - OTS (available now)	
Application-specific - OTS (in development)	
Not Available	

Last update 31-Jan-06

SC Availability SMIC

ARM Standard Cell Product Availability for SMIC

	90nm		130nm		150nm	180nm
STANDARD CELL PRODUCTS	G	LL	G	LL	SMC	SMC
SAGE/SAGE-X (speed & density optimized)			StdVt, HVt			
Metro (low power and high density)		StdVt, HVt	StdVt			
Advantage (high speed & density optimized)	StdVt, HVt					
SAGE-HS (very high speed optimized)						
Advantage-HS (very high speed optimized)						

Explanation	
Free library program (available now)	
Free library program (in development)	
Not Available	

Summary

- ARM standard cell libraries provide a comprehensive feature set to reduce dynamic and leakage power
- Chip designer can select which combination of features are best suited for their design
- ARM works with leading EDA companies to provide automated methodologies to support all power optimization features